AEROTECH ENGINEERING SPECIFICATION: ES16933-2 **REVISION: A**

Customer Name: FERMILAB

Eng. Appd: TES **Order Number:** 185744

Eng. Date: 2/19/2010 **Release Appd:** **AGS**

Subject/Description: NPAQ SYSTEM SPECIFICATION **Release Date:** 10/22/2016

CUST FILENAME: 185744 ES16933-2A1 NPAQ SYSTEM SPECIFICATION.XLS

Hardware Axis Designation:	Axis-1 (X)	Axis-2 (Y)	Axis-3 (Z)
NPAQ Model:	NPAQ-A-80B-80B/ULTRA/NO SPLIT/STAND COOL/STAND/BRAKE-3/ESTOP2/MXR-6/M6		
Axis Amplifier:	DP32030E	DP32020E	DP32020E
Motor Bus Voltage:	+ / - 80 VDC (160 VDC)	+ / - 80 VDC (160 VDC)	+ / - 80 VDC (160 VDC)
Axis Interconnect Drawing(s):			
Motor Cable Type / Port:	C22071-10 ADAPTOR to GANTRY INTEGRAL MOTOR CABLE	C22071-10 ADAPTOR to GANTRY INTEGRAL MOTOR CABLE	C22071-10 ADAPTOR to GANTRY INTEGRAL MOTOR CABLE
Feedback Cable Type / Port:	GANTRY INTEGRAL FEEDBACK CABLE	GANTRY INTEGRAL FEEDBACK CABLE	GANTRY INTEGRAL FEEDBACK CABLE
Stage:	ES13805-1 AGS10000 BASE AXIS	ES13805-1 AGS10000 BRIDGE AXIS	ES13805-1 ATS100-100 VERTICAL AXIS
Nominal Travel:	500 MM	500 MM	100 MM
Limit Type:	NORM CLOSED, OPEN COLL, SINKING	NORM CLOSED, OPEN COLL, SINKING	NORM CLOSED, OPEN COLL, SINKING
Ballscrew / Lead / Gear Ratio:	DIRECT DRIVE	DIRECT DRIVE	2 MM BALLSCREW, 10:1 GEARHEAD
Encoder:	RENISHAW RGH22X (.02MM PITCH, X5 MULT, TTL LINE DRIVEN)	RENISHAW RGH22X (.02MM PITCH, X5 MULT, TTL LINE DRIVEN)	E1000H (TTL LINE DRIVEN)
Encoder Multiplier:	ENC X5, X4 INTERPOLATION = X20 OVERALL	ENC X5, X4 INTERPOLATION = X20 OVERALL	X4 INTERPOLATION
Motor:	DUAL BLM325-A (3.25 OHMS, 2.6MH, 44.88 V/M/S, 39.1 N-M/AMPpk)	BLM264-A (5.3 OHMS, 4.2MH, 35.83 V/M/S, 31.54 N-M/AMPpk)	BM75/ES13805-6 (1 OHM, .8 MH, 9 V/KRPM, .06 N-M/AMPpk)
Motor cooling:	20 psi	20 psi	NONE
Motor I peak / I cont-peak:	30 AMPS / 10 AMPS w AIR	20 AMPS / 8.77 AMPS w AIR	20 AMPS / 9 AMPS
Brake Type:	NONE	NONE	NONE
Number of Poles or Pole Pitch:	30.48 MM POLE PITCH, 60.96 MM CYCLE	30.48 MM POLE PITCH, 60.96 MM CYCLE	8 POLES, 4 CYCLES
Counts / rev :			1000 X 4 = 4,000 CTS/MOTOR REV
Counts / cycle:	60.96 MM / .001 = 60,960 CTS/CYCLE	60.96 MM / .001 = 60,960 CTS/CYCLE	4000 / 4 = 1,000 CTS/CYCLE
Machine Resolution / User Units:	.02MM / 5 / 4 = .001 MM	.02MM / 5 / 4 = .001 MM	2MM / 10 / 4000 = .00005 MM
Programming Resolution / User Units:	.001 MM	.001 MM	.0001 MM
User Programming Units:	MM & MM / SEC	MM & MM / SEC	MM & MM / SEC
Max. Speed Counts / Sec:	1,000,000 CTS / SEC	1,000,000 CTS / SEC	300,000 CTS/SEC
Max. Speed User Units / Sec:	1000 MM / SEC	1000 MM / SEC	15 MM / SEC
Max. Speed Limiting Factor:	USER MAX SPEED SPEC	USER MAX SPEED SPEC	USER MAX SPEED SPEC
Acceleration Rate Units / Sec²:	2500 MM / SEC / SEC	2500 MM / SEC / SEC	150 MM/ SEC / SEC
Acceleration Ramp Type:	SINUSOIDAL	SINUSOIDAL	SINUSOIDAL
(+) Program Direction:	CW/+ (AWAY FROM CABLE EGRESS)	CW/+ (AWAY FROM CABLE EGRESS)	CCW/- (CARRIAGE UP)
Home Direction:	CCW/- (TOWARDS CABLE EGRESS)	CCW/- (TOWARDS CABLE EGRESS)	CCW/- (CARRIAGE UP)
Home Location:	1ST MARKER FROM CCW/- LIMIT	1ST MARKER FROM CCW/- LIMIT	1ST MARKER FROM CCW/- LIMIT
Test Axis loads:			2 KG
Axis Special Test(s):			
System Parameter File Name:	185744-A-1-1.prma		
Axis Calibration File Name:			

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Notes:

1) THE ES13805-1 GANTRY ASSEMBLY IS FROM SO#117468.

*** Axis "Machine Direction" is defined by Motor mounting orientation.**

*** Axis "Program Direction" may or may not be the same as "machine direction" and is defined by the axis application/requirement.**

Hardware Axis Designation:	Axis-4 (U)	Axis-5 (A)	Axis-6 (B)
NPAQ Model:	NPAQ-A-80B-80B/ULTRA/NO SPLIT/STAND COOL/STAND/BRAKE-3/ESTOP2/MXR-6/M6		
Axis Amplifier:	DP32030E	DP32020E	DP32020E
Motor Bus Voltage:	+ / - 80 VDC (160 VDC)	+ / - 80 VDC (160 VDC)	+ / - 80 VDC (160 VDC)
Axis Interconnect Drawing(s):			
Motor Cable Type / Port:	C22071-10 ADAPTOR to GANTRY INTEGRAL MOTOR CABLE		
Feedback Cable Type / Port:	GANTRY INTEGRAL FEEDBACK CABLE, MXC-3		
Stage:	ES13805-1 BMS100 ROTARY AXIS		
Nominal Travel:	360 DEG CONTINUOUS		
Limit Type:	NONE		
Ballscrew / Lead / Gear Ratio:	DIRECT DRIVE		
Encoder:	E1000AS (4 VP-P SINEWAVE)		
Encoder Multiplier:	MXH250/ES13805-5, X4 INTERPOLATION = X1000 OVERALL		
Motor:	BMS100/ES13805-4 (12.9 OHM, 2.4 MH, 40 V/KRPM, .27 N-M/AMPpk)		
Motor cooling:	NONE		
Motor I peak / I cont-peak:	8.4 AMPS / 2.1 AMPS		
Brake Type:	NONE		
Number of Poles or Pole Pitch:	8 POLES, 4 CYCLES		
Counts / rev :	1000 X 250 X 4 = 1,000,000 CTS/MOTOR REV		
Counts / cycle:	1,000,000 / 4 = 250,000 CTS/CYCLE		
Machine Resolution / User Units:	360 DEG / 1,000,000 = .00036 DEGREES		
Programming Resolution / User Units:	.001 DEG		
User Programming Units:	DEG & DEG / SEC		
Max. Speed Counts / Sec:	7,000,000 CTS/SEC		
Max. Speed User Units / Sec:	2520 DEG / SEC		
Max. Speed Limiting Factor:	USER MAX SPEED SPEC		
Acceleration Rate Units / Sec²:	25,200 DEG / SEC / SEC		
Acceleration Ramp Type:	SINUSOIDAL		
(+) Program Direction:	CW/+ (CW MOTOR ROTATION)		
Home Direction:	CCW/- (CCW MOTOR ROTATION)		
Home Location:	AT MARKER		
Test Axis loads:	.00006 KGxM^2		

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Axis Special Test(s):			
System Parameter File Name:	185744-A-1-1.prma		
Axis Calibration File Name:			

Notes:

*** Axis "Machine Direction" is defined by Motor mounting orientation.**

*** Axis "Program Direction" may or may not be the same as "machine direction" and is defined by the axis application/requirement.**